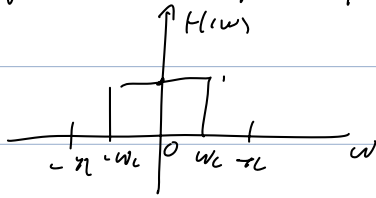


Impulse response from frequency response

for low pass filter:



$$H(w) = \begin{cases} 1, & |w| < w_c \\ 0, & w_c < w \leq \pi \\ & -\pi < w < -w_c \end{cases}$$

Since the impulse response is the DFT of the frequency response, then:

$$h(n) = \frac{1}{2\pi} \cdot \int_{-w_c}^{w_c} e^{jwn} dw$$

* when $n = 0$, then:

$$h(0) = \frac{1}{2\pi} \cdot w \Big|_{-w_c}^{w_c} = \frac{w_c}{\pi}$$

* when $n \neq 0$, then:

$$h(n) = \frac{1}{2\pi} \cdot \frac{1}{jn} \cdot e^{jwn} \Big|_{-w_c}^{w_c}$$

$$= \frac{1}{2\pi} \cdot (e^{jw_c n} - e^{-jw_c n}) / jn$$

$$= \frac{w_c}{\pi} \frac{\sin w_c n}{w_c n}$$

$$= \frac{w_c}{\pi} \text{sinc}(w_c n)$$

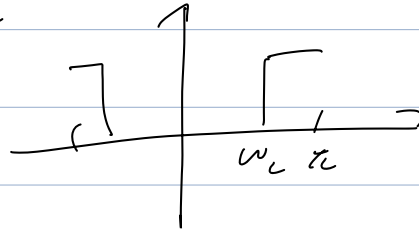
$$\sin \theta = \frac{e^{j\theta} - e^{-j\theta}}{2j}$$

for high pass filter, the freq. response:

$$H(\omega) = \begin{cases} 0 & -\pi < \omega < \omega_c \\ 1 & \omega_c < \omega < \pi \end{cases}$$

By performing ZFT:

$$h(n) = \frac{1}{2\pi} \int_{\omega_c}^{\pi} e^{j\omega n} d\omega$$



$$+ \frac{1}{2\pi} \int_{-\pi}^{-\omega_c} e^{j\omega n} d\omega$$

* When $n=0$, $h(0) = \frac{\pi - \omega_c}{2\pi} + \frac{-\omega_c + \pi}{2\pi} = \frac{2\pi - 2\omega_c}{2\pi} = 1 - \frac{\omega_c}{\pi}$

* When $n \neq 0$, $h(n) = \frac{1}{2\pi} \cdot \frac{e^{j\pi n} - e^{j\omega_c n}}{jn} + \frac{1}{2\pi} \cdot \frac{e^{-j\omega_c n} - e^{-j\pi n}}{jn}$

$$= \frac{1}{2\pi} \cdot \frac{-(e^{j\omega_c n} - e^{-j\omega_c n})}{jn}$$

$$= -\frac{\omega_c}{\pi} \cdot \frac{\sin \omega_c n}{\omega_c n}$$